11 (a) (i) unwanted random power / signal / energy B1 [1]

(ii) loss of (signal) power / energy B1 [1]

(b) (i) *either* signal-to-noise ratio at mic. $= 10\,\lg\,(P_2 / P_1)$ C1

$= 10\,\lg\,(\{2.9 \times 10^{-6}\} / \{3.4 \times 10^{-9}\})$

$= 29\,\text{dB}$ A1

maximum length $= (29 - 24) / 12$ C1

$= 0.42\,\text{km} = 420\,\text{m}$ A1 [4]

or signal-to-noise ratio at receiver $= 10\ \lg\,(P_2 / P_1)$ (C1)

at receiver, $24 = 10\ \lg(P / \{3.4 \times 10^{-9}\})$

$P = 8.54 \times 10^{-7}\ \text{W}$ (A1)

power loss in cables $= 10\,\lg(\{2.9 \times 10^{-6}\} / \{8.54 \times 10^{-7}\})$ (C1)

$= 5.3\,\text{dB}$

length $= 5.3 / 12\,\text{km}$

$= 440\,\text{m}$ (A1)

(ii) use an amplifier M1
coupled to the microphone A1 [2]
(*repeater amplifiers scores no mark*)

12 **(a)** (carrier wave) transmitted from Earth to satellite (1)